

Frequency-Domain Deblurring

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Implementing and comparing Fourier-domain restoration methods against spatial-domain baselines for deblurring

Degradation model

$$g(x, y) = h(x, y) * f(x, y) + n(x, y) \quad (1)$$

$$G(u, v) = H(u, v) F(u, v) + N(u, v) \quad (2)$$

f = sharp image, h = blur kernel (PSF), n = additive noise, g = observed blurry image; capitals denote the corresponding 2-D discrete Fourier transforms, and $\hat{F}$ denotes the restored estimate of F .

Blur kernels

$$h_{\text{motion}}(x, y; L, \theta) = \frac{1}{Z} \delta(x \cos \theta - y \sin \theta) \mathbb{1}_{\left[\sqrt{x^2 + y^2} \leq \frac{L}{2} \right]} \quad (3)$$

$$h_{\text{gauss}}(x, y; \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (4)$$

normalised so that $\sum_{x,y} h(x, y) = 1$. The transfer function is obtained by zero-padding the PSF to the image size, circularly centring it, and transforming:

$$H(u, v) = \mathcal{F}\{h_{\text{pad}}(x, y)\} \quad (5)$$

Inverse filter

$$\hat{F}(u, v) = \frac{G(u, v)}{H_\varepsilon(u, v)}, \quad H_\varepsilon = \begin{cases} H, & |H| \geq \varepsilon, \\ \varepsilon, & |H| < \varepsilon, \end{cases} \quad \varepsilon = 10^{-2} \quad (6)$$

Wiener filter

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + K} G(u, v), \quad K = 10^{-2} \quad (7)$$

H^* = complex conjugate; K approximates the noise-to-signal power ratio S_n/S_f .

Constrained least squares (Laplacian regularisation)

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + \gamma |P(u, v)|^2} G(u, v), \quad \gamma = 10^{-2} \quad (8)$$

where $P = \mathcal{F}\{p\}$ is the transform of the Laplacian operator

$$p = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (9)$$

Spatial baseline: unsharp mask

$$\hat{f}(x, y) = g(x, y) + \lambda \left[g(x, y) - (G_\sigma * g)(x, y) \right], \quad \sigma = 2, \quad \lambda = 1.5 \quad (10)$$

Spatial baseline: median filter

$$\hat{f}(x, y) = \underset{(i,j) \in \mathcal{N}_{3 \times 3}(x,y)}{\text{median}} \, g(i, j) \quad (11)$$

Restoration

$$\hat{f}(x, y) = \mathcal{F}^{-1} \left\{ \hat{F}(u, v) \right\}, \quad \hat{f} \leftarrow \text{clip} \left(\hat{f}, 0, 1 \right) \quad (12)$$

Evaluation metrics

$$\text{MSE} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \left[\hat{f}(x, y) - f(x, y) \right]^2 \quad (13)$$

$$\text{PSNR} = 10 \log_{10} \left(\frac{L^2}{\text{MSE}} \right), \quad L = 1 \quad (14)$$

$$\text{SSIM}(\hat{f}, f) = \frac{\left(2\mu_{\hat{f}}\mu_f + C_1 \right) \left(2\sigma_{\hat{f}f} + C_2 \right)}{\left(\mu_{\hat{f}}^2 + \mu_f^2 + C_1 \right) \left(\sigma_{\hat{f}}^2 + \sigma_f^2 + C_2 \right)} \quad (15)$$

with $C_1 = (K_1 L)^2$, $C_2 = (K_2 L)^2$, $K_1 = 0.01$, $K_2 = 0.03$.

Kernel selection

$$\left(\hat{L}, \hat{\theta} \right) = \arg \max_{L \in \mathcal{L}, \theta \in \Theta} \text{PSNR} \left(\hat{f}(g; h_{L, \theta}), f \right) \quad (16)$$

$$\mathcal{L} = \{5, 9, 13, 17, 21\}, \quad \Theta = \{0^\circ, 30^\circ, 60^\circ, 90^\circ, 120^\circ, 150^\circ\}, \quad \sigma \in \{1, 1.5, 2, 3\} \quad (17)$$